to supply nutrients to all parts of the body. This continuous activity uplifts the role of the heart as a vital organ.

Blood pumping cycle of the heart is called as cardiac cycle, which ensure that the blood is distributed throughout the body. The oxygen distribution process begins when oxygen –free blood (impure) enters in to the heart through the right atrium goes in to the right ventricle, enters the lungs for oxygenation and release of carbondi-oxide and transfers in to left atrial chambers, ready for re-distribution. About **5-6 liters** of blood circulates in the body and cardiac cycles are completed per minute.

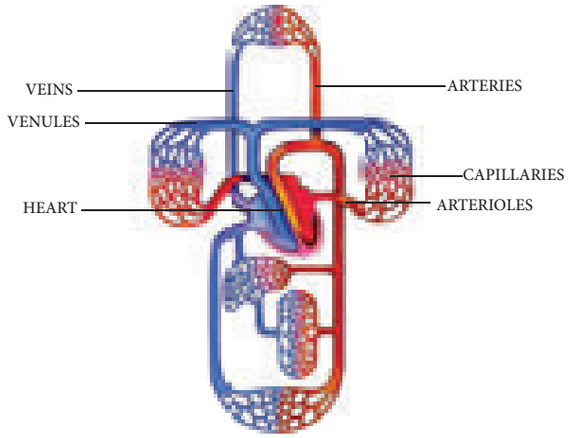
DO YOU KNOW?

A healthy heart beats 70 to 75 times per minute, each cardiac cycle, or heartbeat, takes about 0.8 seconds to complete the cycle.

Oxygen reloading process occur in two phase. The **systole** is a short period that occurs, when the tricuspid and mitral valve close.

The **diastole** is a relatively longer period when the aortic and pulmonary valve close. The systole –disastole relationship is the **mean** blood pressure. Other ways of physically determining the regular functioning of the heart is through examining the pulse rate.

Circulatory System



Circulatory system

In the fig - 1.6 shows Red color indicates oxygenated blood carried in arteries. Blue indicates deoxygenated blood carried in veins.

The relationship between the heart and different types of blood vessels.

Heart

- Aorta
- Arteries
- Arterioles
- Capillaries
- Venules
- Veins
- Venacava
- Heart

Arteries

Arteries are the blood vessels that carries oxygenated blood through out the body. Arteries consists of several layers and smooth muscles that enable them to pump blood throughout the body after it leaves the heart.

ARTERIES - It has three layers

- Tunica adventica
- Tunica media
- Tunica intima

Arterioles

Blood valve that receives blood from the arteries. Those are present next to the arteries and before the capillaries. Arterioles also have smooth muscles.

Capillaries

These are the smallest structure of the circulatory system. The point at which the exchange of oxygen and corbon di-oxide occurs through the thin walls of the capillaries.

Venules

Blood vessels that receive blood from the capillaries and transport deoxygenated blood to the veins.

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